

WASP-52b

R-1783

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-52_200901 · created 2026-08-02T23:50:55+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459093.7681 +/- 0.0056 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.093 +/- 0.063
Transit depth (Rp/Rs)^2	0.86 +/- 1.16 [%]
Orbital Inclination (inc)	80.47 +/- 2.82
Airmass coefficient 1 (a1)	0.59 +/- 0.43
Airmass coefficient 2 (a2)	0.4 +/- 0.78
Scatter in the residuals of the lightcurve fit is	69.45 %
Best Comparison Star	#1 - [567, 48]
Optimal Aperture	3.0
Optimal Annulus	5.5
Transit Duration (day)	0.017 +/- 0.025

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.093 ± 0.063 vs 0.1646 (deviation 1.14σ)
Duration fit vs published	0.017 d vs 0.0758 d (deviation 2.35σ)
Mid-time O–C	-1.1 min
Depth SNR	1.5
Residual scatter	69.45 %

Light curve

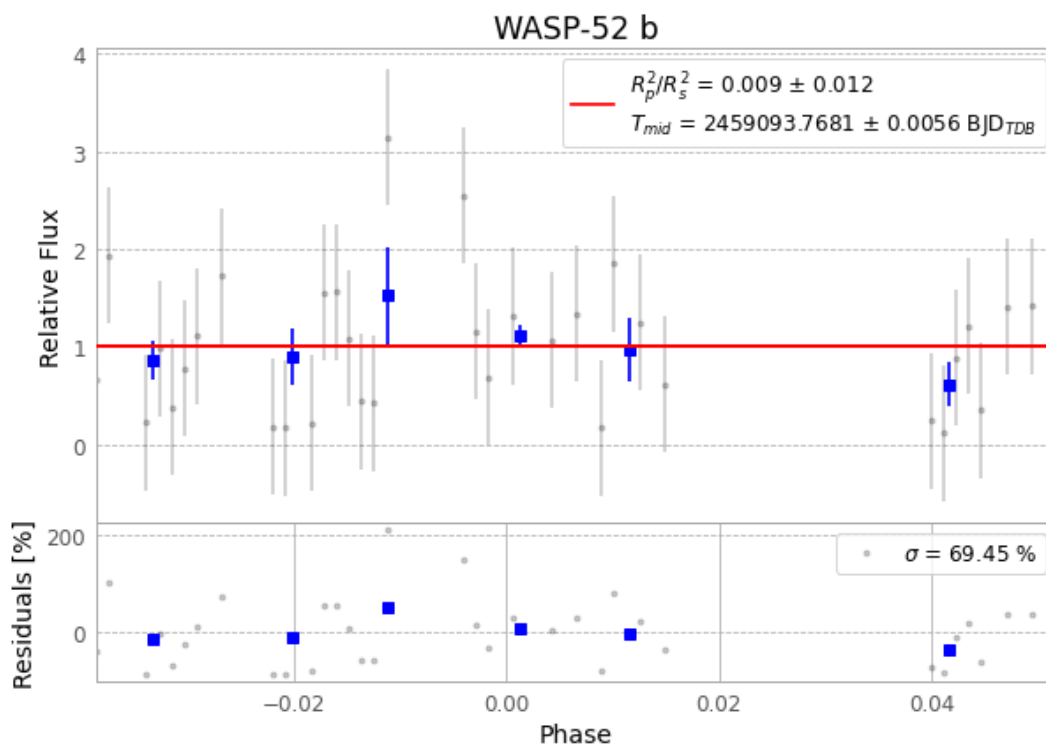


Figure 1 — FinalLightCurve_WASP-52 b_2020-08-31.png (SHA-256 7831c466bbce9861...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [360, 438], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2074616d928f34ea...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

77 FITS frames (51 MB), WASP-52200901043912.FITS → WASP-52200901082712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-52-200901.log (31196b8029a2...), FinalLightCurve_WASP-52 b_2020-08-31.png (7831c466bbce...), FinalLightCurve_WASP-52 b_2020-08-31.csv (d2063b5619e8...)

Compute resources

Wall time 140.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 23:50Z.